

## 32.3 Prediction: BLUP and Henderson's equations

The twelve laboratories of Example (A laboratory proficiency study) each have a realized effect  $a_k$ . It is not a parameter but the value taken by a random variable, so asking for it is a prediction problem. The answer is worth the trouble of setting the problem up properly: it is not the laboratory's own average, and the difference is shrinkage of exactly the kind [Chapter 27](#) studied, arriving here without anybody having decided to be biased.

### 32.3.1 The criterion

#### Definition 32.3.1. Mixed functions and linear predictors

In Definition 32.2.1, a **mixed function** is a random variable  $\tau = \lambda^\top \beta + \mathbf{c}^\top \mathbf{u}$  with known  $\lambda \in \mathbb{R}^p$  and  $\mathbf{c} \in \mathbb{R}^q$ . A **linear predictor** of  $\tau$  is a statistic  $t = \mathbf{a}^\top \mathbf{Y} + b$  with known  $\mathbf{a}$  and  $b$ ; it is **unbiased** if  $E(t) = E(\tau)$  for every  $\beta \in \mathbb{R}^p$ . The **best linear unbiased predictor** (BLUP) of  $\tau$  is the unbiased linear predictor minimizing the prediction mean squared error  $E(t - \tau)^2$ .

Since  $E(t) = \mathbf{a}^\top \mathbf{X}\beta + b$  and  $E(\tau) = \lambda^\top \beta$ , unbiasedness for every  $\beta$  forces  $b = 0$  and  $\mathbf{X}^\top \mathbf{a} = \lambda$ . So an unbiased linear predictor exists exactly when  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ , that is, when  $\lambda^\top \beta$  is estimable in the sense of Definition 8.2.1; the random part  $\mathbf{c}^\top \mathbf{u}$  places no restriction at all, because its mean is zero whatever  $\mathbf{c}$  is. Note also that the criterion averages over  $\mathbf{u}$  as well as over  $\epsilon$ . This is the decision that makes the problem solvable, and the warning in the next subsection says what it costs.

The problem has already been solved. Theorem 14.2.8 of [Section 14.2](#) predicted a new response  $y_0 = \mathbf{x}_0^\top \beta + e_0$  whose error was correlated with the training errors, and  $\mathbf{c}^\top \mathbf{u}$  is exactly such an  $e_0$ . What follows is that theorem in mixed-model notation; its proof is the substitution, plus the uniqueness and the decomposition (c) that [Chapter 14](#) did not need.

#### Theorem 32.3.1. Best linear unbiased prediction

In Definition 32.2.1 let  $\mathbf{V} = \mathbf{ZGZ}^\top + \mathbf{R}$  be positive definite and let  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ . Put

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}, \quad \hat{\mathbf{u}} = \mathbf{GZ}^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\hat{\beta}). \quad (32.3.1)$$

- (a) The BLUP of  $\tau = \lambda^\top \beta + \mathbf{c}^\top \mathbf{u}$  is  $\hat{\tau} = \lambda^\top \hat{\beta} + \mathbf{c}^\top \hat{\mathbf{u}}$ . It does not depend on which generalized inverse is used, and it is the unique minimizer.
- (b) Its prediction error variance is

$$E(\hat{\tau} - \tau)^2 = \mathbf{c}^\top (\mathbf{G} - \mathbf{GZ}^\top \mathbf{V}^{-1} \mathbf{ZG}) \mathbf{c} + \mathbf{d}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{d}, \quad (32.3.2)$$

where  $\mathbf{d} = \lambda - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{ZGc}$ .

- (c) If  $\beta$  were known, the best linear predictor of  $\tau$  would be  $\lambda^\top \beta + \mathbf{c}^\top \mathbf{GZ}^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\beta)$ , with prediction error variance the first term of [Eq. 32.3.2](#). The second term is the price of estimating  $\beta$ .

*Proof.* Apply Theorem 14.2.8 to  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + (\mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon})$ , whose error vector has covariance  $\mathbf{V}$ , with  $\sigma^2 = 1$ ,  $y_0 = \tau$ ,  $\mathbf{x}_0 = \boldsymbol{\lambda}$  and  $e_0 = \mathbf{c}^\top \mathbf{u}$ . Since  $\text{Cov}(\mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}, \mathbf{u}) = \mathbf{Z}\mathbf{G}$ , the two covariance inputs of that theorem are

$$\mathbf{v}_0 = \text{Cov}(\mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}, \mathbf{c}^\top \mathbf{u}) = \mathbf{Z}\mathbf{G}\mathbf{c} =: \mathbf{s}, \quad v_{00} = \text{Var}(\mathbf{c}^\top \mathbf{u}) = \mathbf{c}^\top \mathbf{G}\mathbf{c},$$

and its hypothesis  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$  is our  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ . So Eq. 14.2.3 reads  $\hat{\tau} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{c}^\top \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{c}^\top \hat{\mathbf{u}}$ , and its mean squared error is Eq. 32.3.2, because the vector called  $\mathbf{c}$  there is  $\mathbf{x}_0 - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{v}_0 = \boldsymbol{\lambda} - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z}\mathbf{G}\mathbf{c} = \mathbf{d}$  here. That is (a) and (b) apart from the two uniqueness claims.

For those, recall that an unbiased linear predictor is  $t = \mathbf{a}^\top \mathbf{Y}$  with  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ , and that  $t - \tau$  then has mean zero, so

$$\text{E}(t - \tau)^2 = \text{Var}(\mathbf{a}^\top \mathbf{Y} - \mathbf{c}^\top \mathbf{u}) = \mathbf{a}^\top \mathbf{V}\mathbf{a} - 2\mathbf{a}^\top \mathbf{s} + \mathbf{c}^\top \mathbf{G}\mathbf{c}.$$

The last term does not involve  $\mathbf{a}$  and  $\mathbf{V}$  is positive definite, so this is a strictly convex function of  $\mathbf{a}$  on the affine set  $\{\mathbf{a} : \mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}\}$ , non-empty because  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ , and a strictly convex function has at most one minimizer on a convex set. So  $\mathbf{a}$ , and with it  $\hat{\tau}$ , is unique, by an argument that nowhere asks  $\mathbf{X}^\top$  to have full row rank. The generalized inverse then cancels, as one can also see directly:  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$  is the projection of  $\mathbf{Y}$  onto  $\mathcal{C}(\mathbf{X})$  in the  $\mathbf{V}^{-1}$  inner product (Definition 6.9.1), and both  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{r}^\top \mathbf{X}\hat{\boldsymbol{\beta}}$ , for  $\mathbf{r}$  with  $\boldsymbol{\lambda} = \mathbf{X}^\top \mathbf{r}$ , and  $\hat{\mathbf{u}}$  involve  $\hat{\boldsymbol{\beta}}$  only through  $\mathbf{X}\hat{\boldsymbol{\beta}}$ .

For (c), with  $\boldsymbol{\beta}$  known the problem is unconstrained prediction of  $\mathbf{c}^\top \mathbf{u}$  from  $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$ , whose solution is the best linear predictor of Theorem 2.5.2:  $\text{Cov}(\mathbf{c}^\top \mathbf{u}, \mathbf{Y})\mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$ , with error variance  $\mathbf{c}^\top \mathbf{G}\mathbf{c} - \mathbf{s}^\top \mathbf{V}^{-1} \mathbf{s}$ , which is the first term of Eq. 32.3.2 because  $\mathbf{s} = \mathbf{Z}\mathbf{G}\mathbf{c}$ .

■

Two readings of Eq. 32.3.1 are worth having. First,  $\hat{\mathbf{u}}$  is the best linear predictor of  $\mathbf{u}$  with  $\boldsymbol{\beta}$  replaced by its generalized least squares estimate; the whole content of “BLUP” is that this substitution is optimal, not merely convenient. Second, if the model is normal then  $\text{E}(\mathbf{u} | \mathbf{Y}) = \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$  by Theorem 3.5.2, so the BLUP is the conditional mean with  $\boldsymbol{\beta}$  estimated. That is the door into Section 32.6.

### 32.3.2 Shrinkage

Specialize to the balanced one-way model of Definition 32.1.1, with  $\mathbf{X} = \mathbf{1}$ ,  $\mathbf{Z}$  the group indicators,  $\mathbf{G} = \sigma_a^2 \mathbf{I}_g$  and  $\mathbf{R} = \sigma^2 \mathbf{I}$ . By Eq. 32.1.2,  $\mathbf{V}^{-1} = \lambda_1^{-1} \mathbf{M} + \sigma^{-2}(\mathbf{I} - \mathbf{M})$  with  $\lambda_1 = \sigma^2 + m\sigma_a^2$ , and since  $\mathbf{M}\mathbf{Z} = \mathbf{Z}$  we have  $\mathbf{Z}^\top \mathbf{V}^{-1} = \lambda_1^{-1} \mathbf{Z}^\top$ . Also  $\mathbf{1} \in \mathcal{C}(\mathbf{Z})$ , so the generalized least squares estimate of  $\mu$  is the ordinary mean  $\bar{y}$  (Example (Cluster-constant regressors)). Hence

$$\hat{a}_k = \frac{m\sigma_a^2}{\sigma^2 + m\sigma_a^2} (\bar{y}_k - \bar{y}) = B (\bar{y}_k - \bar{y}), \quad B = \frac{m\gamma}{1 + m\gamma}, \quad (32.3.3)$$

with  $\gamma = \sigma_a^2/\sigma^2$ . The predicted group mean is  $\bar{y} + B(\bar{y}_k - \bar{y})$ : the group's own average pulled towards the overall average by  $B \in (0, 1)$ . The pull is strong when groups are small or when groups genuinely differ little, and it vanishes as either  $m$  or  $\gamma$  grows. Formula Eq. 32.3.3 is the empirical Bayes shrinkage of Proposition 27.5.4 with the prior variance supplied by the model, and it is a ridge estimate of the group effects with penalty  $\sigma^2/\sigma_a^2$  (Theorem 27.2.2); the next

subsection makes that identification exact.

*Warning.* Unbiasedness in [Definition 32.3.1](#) averages over  $\mathbf{u}$ . Conditionally on the realized effects it fails, and deliberately so: from [Eq. 32.3.3](#),  $E(\hat{a}_k | a_1, \dots, a_g) = B(a_k - \bar{a})$ , which is nearer zero than  $a_k$ . Predicted effects are *shrunk*, so their spread understates the spread of the true effects, and ranking groups by  $\hat{a}_k$  is not the same as ranking them by  $a_k$ . A league table of hospitals or schools built from BLUPs will move small units towards the middle, which is right for predicting each unit's next observation and wrong if the question is "which unit is genuinely worst".

The shrinkage is paid for in mean squared error. Continuing the balanced one-way case, [Eq. 32.3.2](#) with  $\lambda = \mathbf{0}$  and  $\mathbf{c} = \mathbf{e}_k$  gives, after substituting  $\mathbf{G} - \mathbf{GZ}^\top \mathbf{V}^{-1} \mathbf{ZG} = \sigma_a^2 \sigma^2 \lambda_1^{-1} \mathbf{I}$  and  $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} = n/\lambda_1$ ,

$$E(\hat{a}_k - a_k)^2 = \frac{\sigma_a^2}{\lambda_1} \left( \sigma^2 + \frac{m\sigma_a^2}{g} \right), \quad (32.3.4)$$

whereas the obvious predictor  $\bar{y}_k - \bar{y}$  has mean squared error  $\sigma^2(1 - 1/g)/m + \sigma_a^2/g$  ([Exercise \(B1\)](#)).

#### Example. Shrinking twelve laboratories

In Example (A laboratory proficiency study),  $\hat{\sigma}^2 = 0.0620$  and  $\hat{\sigma}_a^2 = 0.0386$  give a shrinkage factor  $B = 0.714$ . The laboratory deviations  $\bar{y}_k - \bar{y}$  run from  $-0.450$  to  $0.250$ ; the predicted effects run from  $-0.321$  to  $0.178$ . Each has been pulled about 29 per cent of the way towards zero.

The gain is real. By [Eq. 32.3.4](#) the prediction error variance is 0.01336, a standard error of 0.1156, against 0.01743 for the raw deviation, a ratio of 1.30 in mean squared error. Panel (b) of [Figure 32.3.1](#) shows the comparison over the whole range of intraclass correlations: shrinkage helps most when the correlation is small, which is exactly when the raw deviations are mostly noise.

### 32.3.3 Henderson's mixed model equations

Formula [Eq. 32.3.1](#) asks for  $\mathbf{V}^{-1}$ , an  $n \times n$  inverse. Henderson (1950) found a system of size  $p + q$  that delivers the generalized least squares estimate and the BLUP together, without ever forming  $\mathbf{V}$ .

#### Theorem 32.3.2. Henderson's mixed model equations

Let  $\mathbf{G}$  and  $\mathbf{R}$  be positive definite and  $\text{rank}(\mathbf{X}) = p$ . Define

$$\mathbf{C} = \begin{pmatrix} \mathbf{X}^\top \mathbf{R}^{-1} \mathbf{X} & \mathbf{X}^\top \mathbf{R}^{-1} \mathbf{Z} \\ \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{X} & \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z} + \mathbf{G}^{-1} \end{pmatrix}, \quad \mathbf{h} = \begin{pmatrix} \mathbf{X}^\top \mathbf{R}^{-1} \mathbf{y} \\ \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{y} \end{pmatrix}. \quad (32.3.5)$$

- (a)  $\mathbf{C}$  is positive definite, and the unique solution of  $\mathbf{C}(\mathbf{b}^\top, \mathbf{v}^\top)^\top = \mathbf{h}$  is  $\mathbf{b} = \hat{\boldsymbol{\beta}}$ , the generalized least squares estimate, and  $\mathbf{v} = \hat{\mathbf{u}}$ , the BLUP of [Eq. 32.3.1](#).

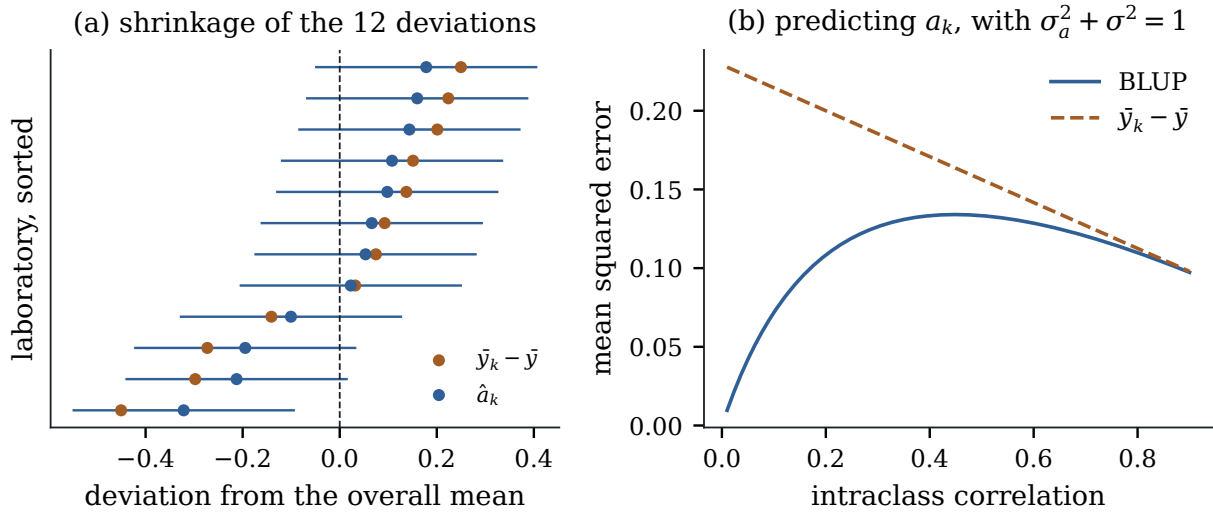


Figure 32.3.1: (a) The twelve laboratory deviations  $\bar{y}_k - \bar{y}$  and their predicted effects  $\hat{a}_k$ , sorted, with 95% prediction intervals from Eq. 32.3.4. (b) Mean squared error of the two predictors of  $a_k$ , as a function of the intraclass correlation, with  $\sigma_a^2 + \sigma^2 = 1$ ,  $g = 12$  and  $m = 4$ .

(b) The same pair is the unique minimizer of the penalized least squares criterion

$$Q(\boldsymbol{\beta}, \mathbf{u}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})^\top \mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) + \mathbf{u}^\top \mathbf{G}^{-1}\mathbf{u}. \quad (32.3.6)$$

(c) With  $\mathbf{W} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}$ , the blocks of  $\mathbf{C}^{-1}$  are

$$(\mathbf{C}^{-1})_{11} = \mathbf{W}^{-1}, \quad (\mathbf{C}^{-1})_{12} = -\mathbf{W}^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G},$$

$$(\mathbf{C}^{-1})_{22} = \mathbf{G} - \mathbf{G} \mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G} + \mathbf{G} \mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{W}^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G}.$$

Its leading block is  $\text{Cov}(\hat{\boldsymbol{\beta}})$  and its trailing block is  $\text{Cov}(\hat{\mathbf{u}} - \mathbf{u})$ , the matrix whose quadratic forms are Eq. 32.3.2.

*Proof.* We use repeatedly the *push-through identity*

$$(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R}^{-1} = \mathbf{G} \mathbf{Z}^\top \mathbf{V}^{-1}, \quad (32.3.7)$$

proved by multiplying on the right by  $\mathbf{V} = \mathbf{R} + \mathbf{Z} \mathbf{G} \mathbf{Z}^\top$ :

$$\begin{aligned} & (\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z})^{-1} (\mathbf{Z}^\top + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z} \mathbf{G} \mathbf{Z}^\top) \\ &= (\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z})^{-1} (\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z}) \mathbf{G} \mathbf{Z}^\top \\ &= \mathbf{G} \mathbf{Z}^\top. \end{aligned}$$

(a) For  $(\mathbf{b}^\top, \mathbf{v}^\top) \neq \mathbf{0}$ ,

$$(\mathbf{b}^\top, \mathbf{v}^\top) \mathbf{C} (\mathbf{b}^\top, \mathbf{v}^\top)^\top = \|\mathbf{R}^{-1/2}(\mathbf{X}\mathbf{b} + \mathbf{Z}\mathbf{v})\|^2 + \mathbf{v}^\top \mathbf{G}^{-1} \mathbf{v},$$

which vanishes only if  $\mathbf{v} = \mathbf{0}$  and  $\mathbf{X}\mathbf{b} = \mathbf{0}$ , hence only if  $\mathbf{b} = \mathbf{0}$  by full column rank. So  $\mathbf{C}$  is positive definite and the system has a unique solution. The second block row

gives

$$\begin{aligned}\mathbf{v} &= (\mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z} + \mathbf{G}^{-1})^{-1} \mathbf{Z}^\top \mathbf{R}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) \\ &= \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b})\end{aligned}$$

by Eq. 32.3.7. Substituting into the first block row, and using the residual identity

$$\begin{aligned}\mathbf{y} - \mathbf{X}\mathbf{b} - \mathbf{Z}\mathbf{v} &= (\mathbf{I} - \mathbf{Z}\mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1}) (\mathbf{y} - \mathbf{X}\mathbf{b}) \\ &= (\mathbf{V} - \mathbf{Z}\mathbf{G}\mathbf{Z}^\top) \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) \\ &= \mathbf{R}\mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}),\end{aligned}$$

the first block row  $\mathbf{X}^\top \mathbf{R}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b} - \mathbf{Z}\mathbf{v}) = \mathbf{0}$  becomes  $\mathbf{X}^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$ , the generalized normal equations of Definition 6.9.1. Hence  $\mathbf{b} = \hat{\boldsymbol{\beta}}$  and then  $\mathbf{v} = \hat{\mathbf{u}}$ .

- (b)  $Q$  is a quadratic in  $(\boldsymbol{\beta}, \mathbf{u})$  with matrix  $\mathbf{C}$  and linear term  $-2\mathbf{h}$ , and the display in (a) shows it is strictly convex. By Proposition 1.11.2 its unique minimizer solves  $\mathbf{C}(\mathbf{b}^\top, \mathbf{v}^\top)^\top = \mathbf{h}$ , which is Eq. 32.3.5, whose solution is  $(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}})$  by part (a).
- (c) Write  $\mathbf{C}_{11} = \mathbf{X}^\top \mathbf{R}^{-1} \mathbf{X}$ ,  $\mathbf{C}_{12} = \mathbf{X}^\top \mathbf{R}^{-1} \mathbf{Z}$  and  $\mathbf{C}_{22} = \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z} + \mathbf{G}^{-1}$ , so that by Eq. 32.2.3 the Schur complement of  $\mathbf{C}_{22}$  is

$$\begin{aligned}\mathbf{C}_{11} - \mathbf{C}_{12} \mathbf{C}_{22}^{-1} \mathbf{C}_{12}^\top &= \mathbf{X}^\top (\mathbf{R}^{-1} - \mathbf{R}^{-1} \mathbf{Z} (\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R}^{-1}) \mathbf{X} \\ &= \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} = \mathbf{W}.\end{aligned}$$

Theorem 1.6.4 now gives  $(\mathbf{C}^{-1})_{11} = \mathbf{W}^{-1}$  and  $(\mathbf{C}^{-1})_{12} = -\mathbf{W}^{-1} \mathbf{C}_{12} \mathbf{C}_{22}^{-1} = -\mathbf{W}^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z}\mathbf{G}$ , the last step being the transpose of Eq. 32.3.7. For the trailing block,

$$\begin{aligned}(\mathbf{C}^{-1})_{22} &= \mathbf{C}_{22}^{-1} + \mathbf{C}_{22}^{-1} \mathbf{C}_{12}^\top \mathbf{W}^{-1} \mathbf{C}_{12} \mathbf{C}_{22}^{-1} \\ &= \mathbf{C}_{22}^{-1} + \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{W}^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z}\mathbf{G},\end{aligned}$$

and  $\mathbf{C}_{22}^{-1} = \mathbf{G} - \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{Z}\mathbf{G}$ , because

$$\begin{aligned}(\mathbf{G} - \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{Z}\mathbf{G})(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z}) &= \mathbf{I} + \mathbf{G}\mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z} - \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} (\mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}) \mathbf{R}^{-1} \mathbf{Z} = \mathbf{I},\end{aligned}$$

using Eq. 32.3.7 once more in the form  $\mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{V} = \mathbf{G}\mathbf{Z}^\top$ . Finally,  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \mathbf{W}^{-1}$  is Corollary 7.2.3, and the trailing block is Eq. 32.3.2 with  $\boldsymbol{\lambda} = \mathbf{0}$  read off entry by entry. ■

*Idea.* Henderson's system says that a mixed model is a penalized least squares problem (Eq. 32.3.6): fit the random effects as though they were free parameters, but charge  $\mathbf{u}^\top \mathbf{G}^{-1} \mathbf{u}$  for using them. With  $\mathbf{R} = \sigma^2 \mathbf{I}$  and  $\mathbf{G} = \sigma_u^2 \mathbf{I}$  the criterion is  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}\|^2 + \lambda \|\mathbf{u}\|^2$  with  $\lambda = \sigma^2/\sigma_u^2$ : ridge regression on the random-effect columns only (Theorem 27.2.2). The difference from Chapter 27 is that the penalty is not chosen by cross-validation; it is estimated, because the model says what  $\sigma^2/\sigma_u^2$  means. Chapter 43 exploits the same equivalence in the other direction, writing a penalized spline as a mixed model so that its smoothing parameter can be estimated by REML (Section 43.6).

```
import numpy as np
```

```

g, m = 12, 4
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25          # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

X = np.ones((n, 1))                            # one fixed effect, the grand mean
Z = np.repeat(np.eye(g), m, axis=0)            # laboratory indicators

Y = y.reshape(g, m)                            # moment estimates, Section 32.1
lab_mean = Y.mean(axis=1)
ms_between = m * np.sum((lab_mean - Y.mean()) ** 2) / (g - 1)
ms_within = np.sum((Y - lab_mean[:, None]) ** 2) / (n - g)
s2, s2a = ms_within, (ms_between - ms_within) / m

G = s2a * np.eye(g)
R = s2 * np.eye(n)
V = Z @ G @ Z.T + R

```

[Run this code online](#)

```

Vinv = np.linalg.inv(V)
beta_gls = np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv @ y)
u_blup = G @ Z.T @ Vinv @ (y - X @ beta_gls)

# Henderson's mixed model equations: one (p + q) x (p + q) system, no n x n inverse
Rinv = np.linalg.inv(R)
C = np.block([[X.T @ Rinv @ X, X.T @ Rinv @ Z],
              [Z.T @ Rinv @ X, Z.T @ Rinv @ Z + np.linalg.inv(G)]])
rhs = np.concatenate([X.T @ Rinv @ y, Z.T @ Rinv @ y])
solution = np.linalg.solve(C, rhs)

print("GLS estimate   ", beta_gls, solution[:1])
print("first two BLUPs", u_blup[:2], solution[1:3])

```

[Run this code online](#)

```

shrinkage = m * s2a / (s2 + m * s2a)            # the shrinkage factor B
print(f"B = {shrinkage:.4f}")
print("u_hat   ", np.round(u_blup[:4], 4))
print("B(ybar_k - ybar)", np.round(shrinkage * (lab_mean - y.mean())[:4], 4))

```

[Run this code online](#)

Two practical points close the section. First,  $\mathbf{C}$  is sparse whenever  $\mathbf{Z}$  is, which is always: each row of  $\mathbf{Z}$  has one non-zero entry per random factor, so Eq. 32.3.5 can be solved by sparse Cholesky factorization, and a model with hundreds of thousands of random effects is routine. Second,  $\mathbf{G}$  and  $\mathbf{R}$  are unknown in practice and estimates are substituted; the result is called the **empirical** BLUP, and Eq. 32.3.2 is then too small, since it ignores the variability of the estimated covariance. Kackar and Harville (1984) identified the extra term, which returns in Section 32.5 as the Kenward–Roger correction and in Section 32.6 as the gap between posterior and plug-in standard errors.

## 32.3.4 Exercises

### 32.3.4.1 A. Check your understanding

#### Exercise. A1

In [Eq. 32.3.3](#), what happens to  $B$  as (a)  $m \rightarrow \infty$ , (b)  $\sigma_a^2 \rightarrow 0$ , (c)  $\sigma^2 \rightarrow 0$ ? Describe in words the predictor in each limit, and say which of the three limits makes the mixed model agree with a fixed-effects analysis.

### 32.3.4.2 B. Practice

#### Exercise. B1

Show that the predictor  $\bar{Y}_k - \bar{Y}$  of  $a_k$  in the balanced one-way model has  $E(\bar{Y}_k - \bar{Y} - a_k)^2 = \sigma^2(1 - 1/g)/m + \sigma_a^2/g$ , and verify that [Eq. 32.3.4](#) is smaller for every  $\gamma > 0$ . Where does the difference come from when  $\gamma$  is very small?

*Solution.*  $\bar{Y}_k - \bar{Y} - a_k = \bar{\varepsilon}_k - \bar{\varepsilon} - \bar{a}$  with  $\bar{a} = g^{-1} \sum_j a_j$ . The three terms have variances  $\sigma^2/m$ ,  $\sigma^2/(gm)$  and  $\sigma_a^2/g$ , and  $\text{Cov}(\bar{\varepsilon}_k, \bar{\varepsilon}) = \sigma^2/(gm)$ , giving  $\sigma^2/m - \sigma^2/(gm) + \sigma_a^2/g$  as stated. In terms of  $\gamma = \sigma_a^2/\sigma^2$ , [Eq. 32.3.4](#) equals  $\sigma^2\gamma(1 + m\gamma/g)/(1 + m\gamma)$  while the raw predictor's error is  $\sigma^2\{(1 - 1/g)/m + \gamma/g\}$ . Putting the two over the denominator  $1 + m\gamma$  and cancelling, their difference is exactly

$$\frac{\sigma^2}{m} \left(1 - \frac{1}{g}\right) \frac{1}{1 + m\gamma} = \frac{\sigma^2}{m} \left(1 - \frac{1}{g}\right) (1 - B),$$

which is strictly positive for every  $\gamma$ . As  $\gamma \rightarrow 0$  the BLUP's error tends to  $\sigma_a^2$  while the raw predictor's tends to  $\sigma^2/m$ , which is much larger: the raw predictor pays the full measurement error even when there is nothing to predict, whereas the BLUP predicts 0 and pays only  $\sigma_a^2$ .

#### Exercise. B2

Take  $\tau = \mu + a_k$ , the mean of group  $k$  in the balanced one-way model, so that  $\lambda = 1$  and  $\mathbf{c} = \mathbf{e}_k$ . Use [Eq. 32.3.1](#) and [Eq. 32.3.2](#) to show that the BLUP is  $\bar{y} + B(\bar{y}_k - \bar{y})$  with prediction error variance  $\sigma^2 B/m + (1 - B)^2 \lambda_1/n$ . Show that this equals  $\lambda_1/n$  at  $B = 0$  and  $\sigma^2/m$  at  $B = 1$ ; that it is strictly below  $\sigma^2/m$ , the prediction error of the group's own mean; and that it is below  $\lambda_1/n$  as well exactly when  $m\sigma_a^2 > \sigma^2(g - 2)$ , so that it is no interpolation between the two — with  $g = 3$ ,  $m = 4$  and  $\sigma^2 = \sigma_a^2 = 1$  it beats both.

*Solution.* The predictor is immediate from [Eq. 32.3.1](#). For the variance,  $\mathbf{d} = \mathbf{1} - \mathbf{1}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G} \mathbf{e}_k = 1 - m\sigma_a^2/\lambda_1 = 1 - B$ , and  $\mathbf{W}^- = \lambda_1/n$ , so the second term of [Eq. 32.3.2](#) is  $(1 - B)^2 \lambda_1/n$ ; the first term is  $\sigma_a^2 \sigma^2/\lambda_1 = \sigma^2 B/m$  since  $B = m\sigma_a^2/\lambda_1$ . Setting  $B = 0$  and  $B = 1$  gives the two stated endpoints. For the comparisons, use  $1 - B = \sigma^2/\lambda_1$  and  $n = gm$  to collapse the whole expression to

$$\frac{\sigma^2}{\lambda_1} \left( \sigma_a^2 + \frac{\sigma^2}{gm} \right),$$

and note that  $\sigma^2/m = (\sigma^2/\lambda_1)(\sigma_a^2 + \sigma^2/m)$ , so the first comparison holds whenever  $g > 1$ . The second asks  $\sigma^2\sigma_a^2gm + \sigma^4 < \lambda_1^2 = (\sigma^2 + m\sigma_a^2)^2$ , which after cancelling  $\sigma^4$  and dividing by  $m\sigma_a^2$  is  $\sigma^2g < 2\sigma^2 + m\sigma_a^2$ . In the numerical case the expression is 13/60, below both 5/12 and 1/4: the quadratic in  $B$  is not an interpolation between two strategies but can beat both, and that gain is what the shrinkage buys.

### Exercise. B3

Take  $\mathbf{X}$  empty (no fixed effects),  $\mathbf{R} = \sigma^2\mathbf{I}$  and  $\mathbf{G} = \sigma_u^2\mathbf{I}$  in Theorem 32.3.2. Show that  $\hat{\mathbf{u}}$  is exactly the ridge estimate  $(\mathbf{Z}^\top\mathbf{Z} + \lambda\mathbf{I})^{-1}\mathbf{Z}^\top\mathbf{y}$  with  $\lambda = \sigma^2/\sigma_u^2$ , and that Eq. 32.3.2 becomes  $\sigma_u^2\mathbf{c}^\top(\mathbf{I} + \lambda^{-1}\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{c}$ . Compare with the mean squared error of ridge in Theorem 27.2.2: which is the same quantity and which is not?

*Solution.* With no  $\mathbf{X}$ , Eq. 32.3.5 reduces to  $(\mathbf{Z}^\top\mathbf{Z}/\sigma^2 + \mathbf{I}/\sigma_u^2)\mathbf{v} = \mathbf{Z}^\top\mathbf{y}/\sigma^2$ , that is  $\mathbf{v} = (\mathbf{Z}^\top\mathbf{Z} + \lambda\mathbf{I})^{-1}\mathbf{Z}^\top\mathbf{y}$ . By Theorem 32.3.2(c) the prediction error covariance is  $\mathbf{C}^{-1} = \sigma^2(\mathbf{Z}^\top\mathbf{Z} + \lambda\mathbf{I})^{-1} = \sigma_u^2(\mathbf{I} + \lambda^{-1}\mathbf{Z}^\top\mathbf{Z})^{-1}$ . The quantity computed here averages over  $\mathbf{u}$ ; the mean squared error in Theorem 27.2.2 is computed for a *fixed*  $\boldsymbol{\beta}$  and is therefore a function of that fixed value. Averaging the ridge mean squared error over  $\boldsymbol{\beta} \sim N(\mathbf{0}, \sigma_u^2\mathbf{I})$  reproduces the BLUP formula, which is the empirical Bayes reading of Proposition 27.5.4.

### 32.3.4.3 C. Going deeper

#### Exercise. C1

Show that the BLUP is invariant to reparameterization of the random effects: if  $\mathbf{Z}^* = \mathbf{Z}\mathbf{T}^{-1}$  and  $\mathbf{G}^* = \mathbf{T}\mathbf{G}\mathbf{T}^\top$  for a nonsingular  $\mathbf{T}$ , with  $\mathbf{u}^* = \mathbf{T}\mathbf{u}$ , then  $\mathbf{V}$  is unchanged and  $\hat{\mathbf{u}}^* = \mathbf{T}\hat{\mathbf{u}}$ . What does this say about predicting a contrast  $\mathbf{c}^\top\mathbf{u}$  directly versus predicting  $\mathbf{u}$  and then forming the contrast?

#### Exercise. C2

Assume the normal mixed model and suppose  $\boldsymbol{\beta}$  is known. Show that among *all* predictors  $t(\mathbf{Y})$ , linear or not, the conditional mean  $E(\mathbf{c}^\top\mathbf{u} \mid \mathbf{Y})$  minimizes  $E(t - \mathbf{c}^\top\mathbf{u})^2$ , and that it coincides with the linear predictor of Theorem 32.3.1(c). Explain why the same argument does not show that  $\hat{\tau}$  is optimal among all predictors when  $\boldsymbol{\beta}$  is unknown.